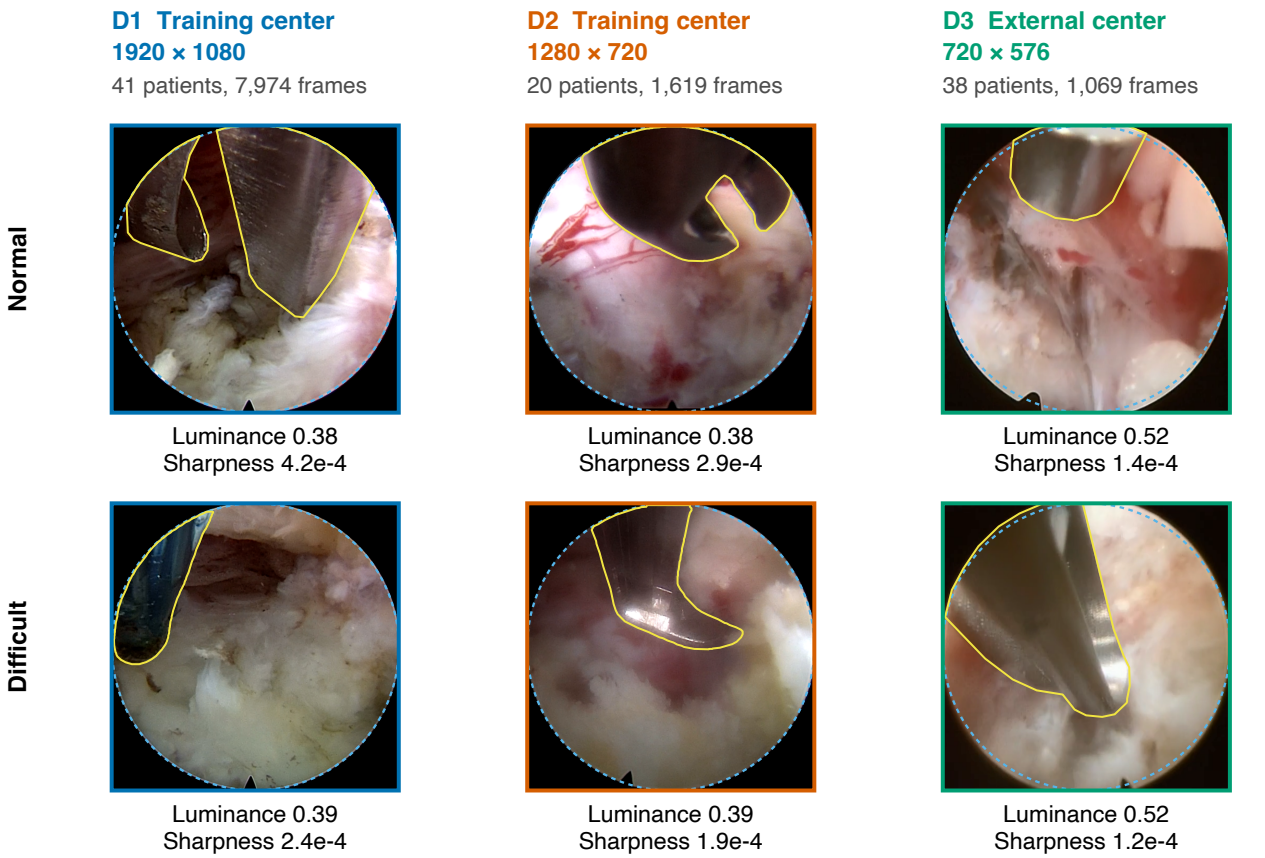
A

SEA DATASET	COHORT AND DOMAINS	ROLE IN THIS STUDY
Annotated frames 99 patients 10,662 frames	**Training center** 61 patients, 9,593 frames 1920 × 1080: 41 patients, 7,974 frames 1280 × 720: 20 patients, 1,619 frames	**5-fold patient-grouped cross-fitting** Out-of-fold masks from two segmenters; optical gate fitted on these, then locked Analyzed: 9,590 frames (3 excluded: no field of view)
		Gate locked before external-center outcomes were analyzed
	External center 38 patients, 1,069 frames 720 × 576: 38 patients, 1,069 frames	**Locked external test** Locked gate; no external-center outcome analyzed before the lock Analyzed: 1,069 frames
Unannotated frames 20 patients 37,848 frames	**Training-center videos, no masks** 1920 × 1080, one frame per second	**Deployment stream** Locked gate; flag rate only Analyzed: 37,823 frames (25 excluded: no field of view)

B



Ground-truth instrument contour

Field-of-view outline

Frame near its group's median luminance and sharpness, instrument covering 4 to 40% of the field of view

Printed values: mean luminance (Rec. 709 luma, 0 to 1) and Tenengrad sharpness (scientific notation)